LAB EXPERIMENTS
Date: 12.08.2026
Questions : 26-30

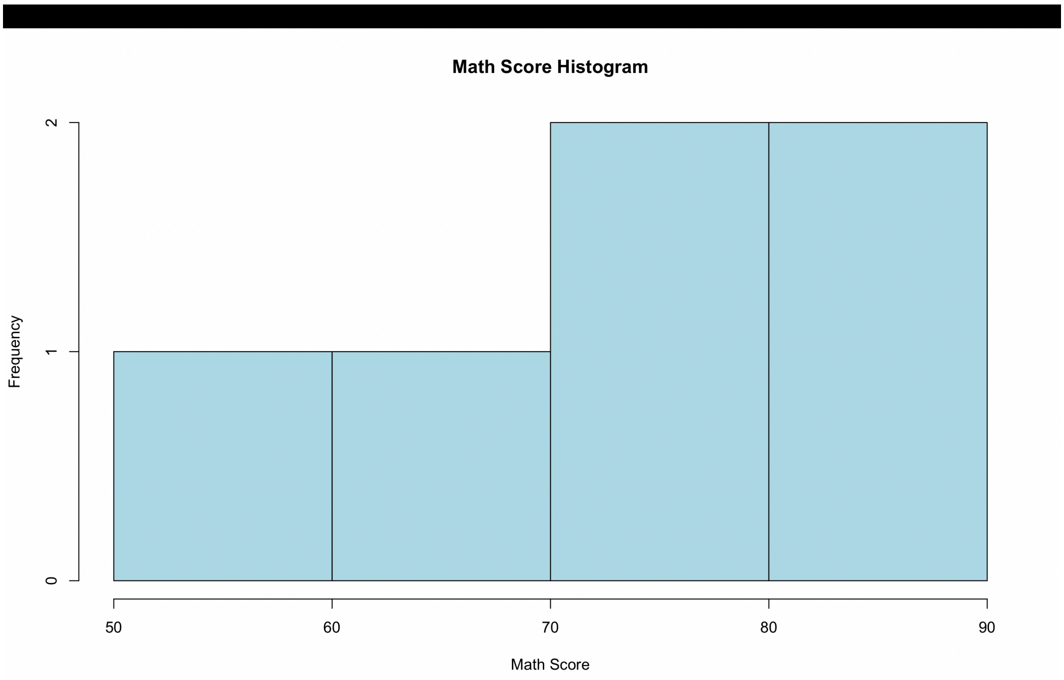
SET 26

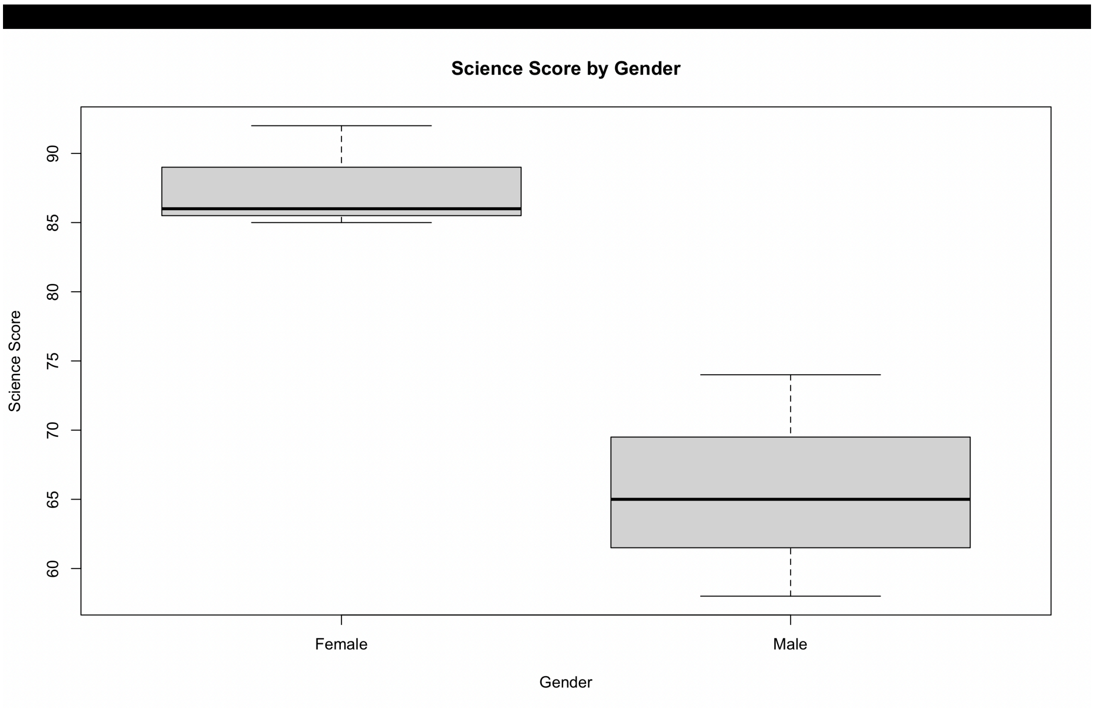
Q1. Histogram of Math Scores and Boxplot of Science Scores by Gender

Program:

```
student <- read.csv("Student_Mini_Data.csv")

hist(student$Math_Score,
     main = "Distribution of Math Scores",
     xlab = "Math Score",
     ylab = "Frequency",
     col = "lightblue",
     border = "black")

boxplot(Science_Score ~ Gender,
        data = student,
        main = "Science Scores by Gender",
        xlab = "Gender",
        ylab = "Science Score",
        col = c("pink", “lightblue"))
```

Output:



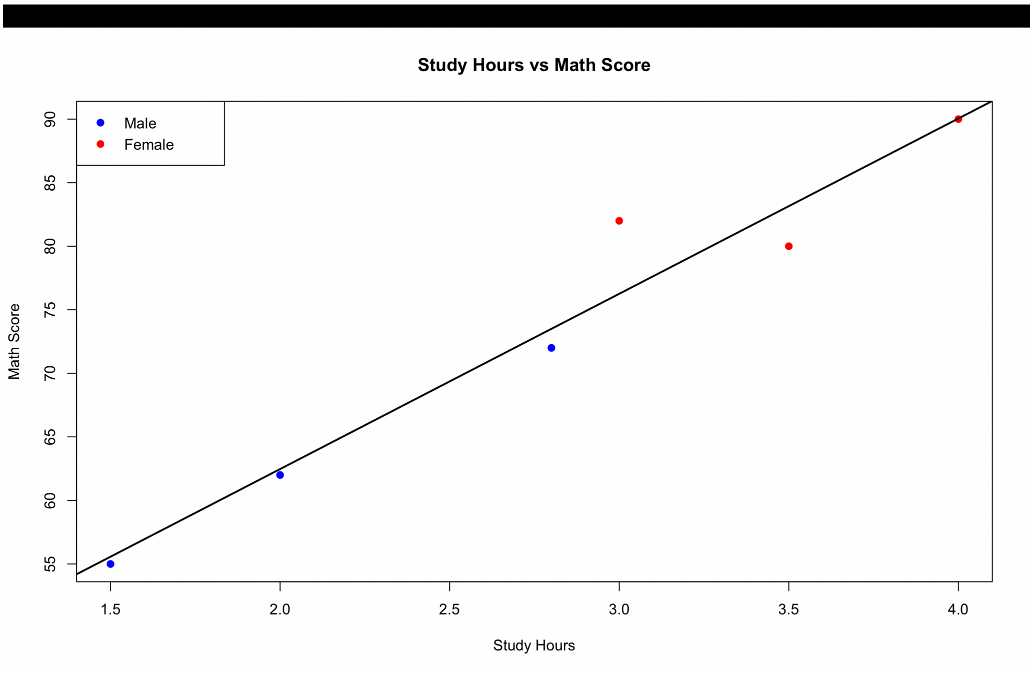


Q2. Scatter Plot of Study Hours and Math Scores with Regression Line

Program:

```
student <- read.csv("Student_Mini_Data.csv")

plot(student$Study_Hours, student$Math_Score,
    col=ifelse(student$Gender=="Male","blue","red"),
    pch=19,
    main="Study Hours vs Math Score",
    xlab="Study Hours",
    ylab="Math Score")

model <- lm(Math_Score ~ Study_Hours, data=student)
abline(model, lwd=2)

legend("topleft",
      legend=c("Male","Female"),
      col=c("blue","red"),
      pch=19)
```

Output:



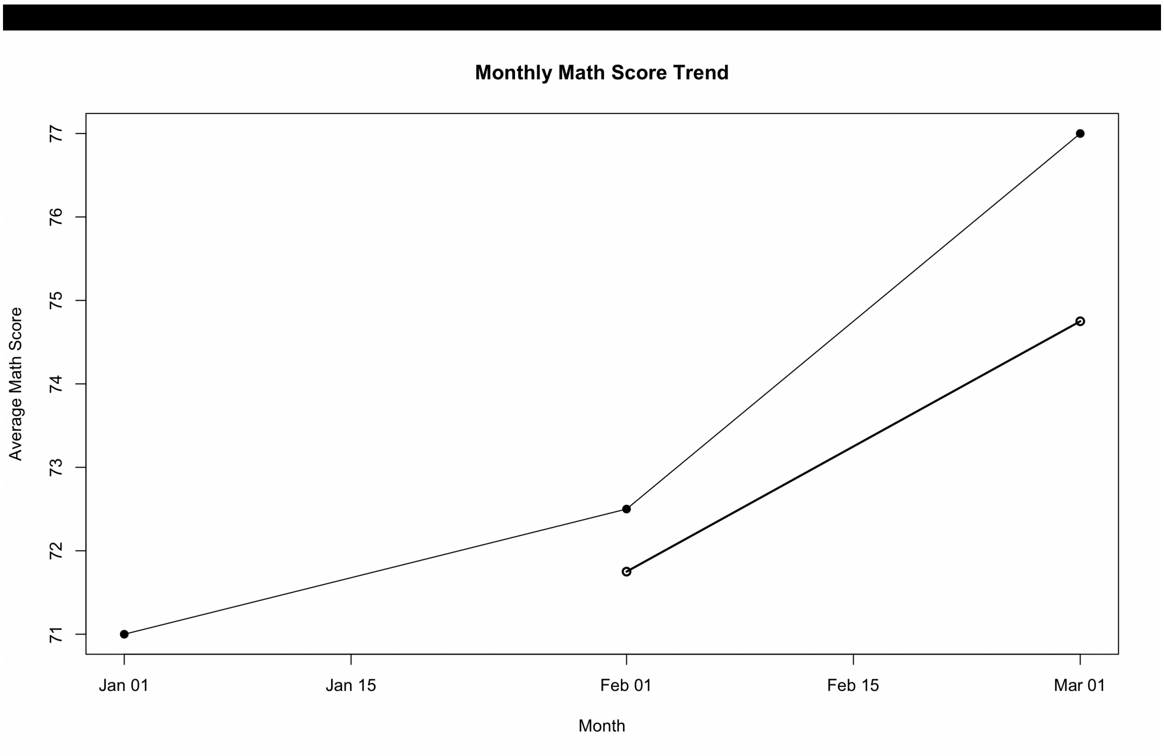
Q3. Monthly Math Score Trend with Moving Average

Program:

```
student <- read.csv("Student_Mini_Data.csv")
student$Exam_Date <- as.Date(student$Exam_Date)
student$Month <- format(student$Exam_Date, "%Y-%m")

monthly_avg <- aggregate(Math_Score ~ Month,
                         data=student,
                         FUN=mean)
print(monthly_avg)

monthly_avg$Month_Date <- as.Date(
  paste0(monthly_avg$Month,"-01")
)

plot(monthly_avg$Month_Date,
     monthly_avg$Math_Score,
     type="o",
     pch=19,
     main="Monthly Math Score Trend",
     xlab="Month",
     ylab="Average Math Score")

monthly_avg$Moving_Average <- filter(
  monthly_avg$Math_Score,
  rep(1/2,2),
  sides=1
```

```
)

lines(monthly_avg$Month_Date,
      monthly_avg$Moving_Average,
      type="o",
      lwd=2)
```

Output:



Q4. Interactive Student Performance Dashboard

Program:

```
# Install packages if needed
install.packages("ggplot2")
install.packages("shiny")

library(ggplot2)
library(shiny)

# Dataset
student <- data.frame(
  Student_ID = c("S01","S02","S03","S04","S05","S06"),
  Gender = c("Male","Female","Male","Female","Male","Female"),
  Study_Hours = c(2.0,3.5,1.5,4.0,2.8,3.0),
  Attendance = c(78,90,70,95,85,92),
```

```
  Math_Score = c(62,80,55,90,72,82),
  Science_Score = c(65,85,58,92,74,86),
  Exam_Date = as.Date(c("2025-01-10","2025-01-10",
                        "2025-02-12","2025-02-12",
                        "2025-03-15","2025-03-15"))
)

# UI
ui <- fluidPage(

  titlePanel("Student Performance Dashboard"),

  sidebarLayout(
    sidebarPanel(
      dateRangeInput("date",
                     "Select Exam Date:",
                     start = min(student$Exam_Date),
                     end = max(student$Exam_Date))
    ),

    mainPanel(
      plotOutput("genderPlot"),
      plotOutput("trendPlot")
    )
  )
)

# Server
server <- function(input, output) {

  filtered_data <- reactive({
    student[student$Exam_Date >= input$date[1] &
              student$Exam_Date <= input$date[2], ]
  })

  # Average Math Score by Gender
  output$genderPlot <- renderPlot({
    ggplot(filtered_data(),
           aes(x = Gender, y = Math_Score, fill = Gender)) +
      stat_summary(fun = mean, geom = "bar") +
      labs(title = "Average Math Score by Gender",
           x = "Gender",
           y = "Average Math Score")
  })

  # Score trend over time
  output$trendPlot <- renderPlot({
    ggplot(filtered_data(),
```

```
      aes(x = Exam_Date, y = Math_Score)) +
    geom_line() +
    geom_point() +
    labs(title = "Math Score Trend Over Time",
      x = "Exam Date",
      y = "Math Score")
  })
}

# Run dashboard
shinyApp(ui = ui, server = server)
```

Output:

SET 27: Patient Health Risk Analysis

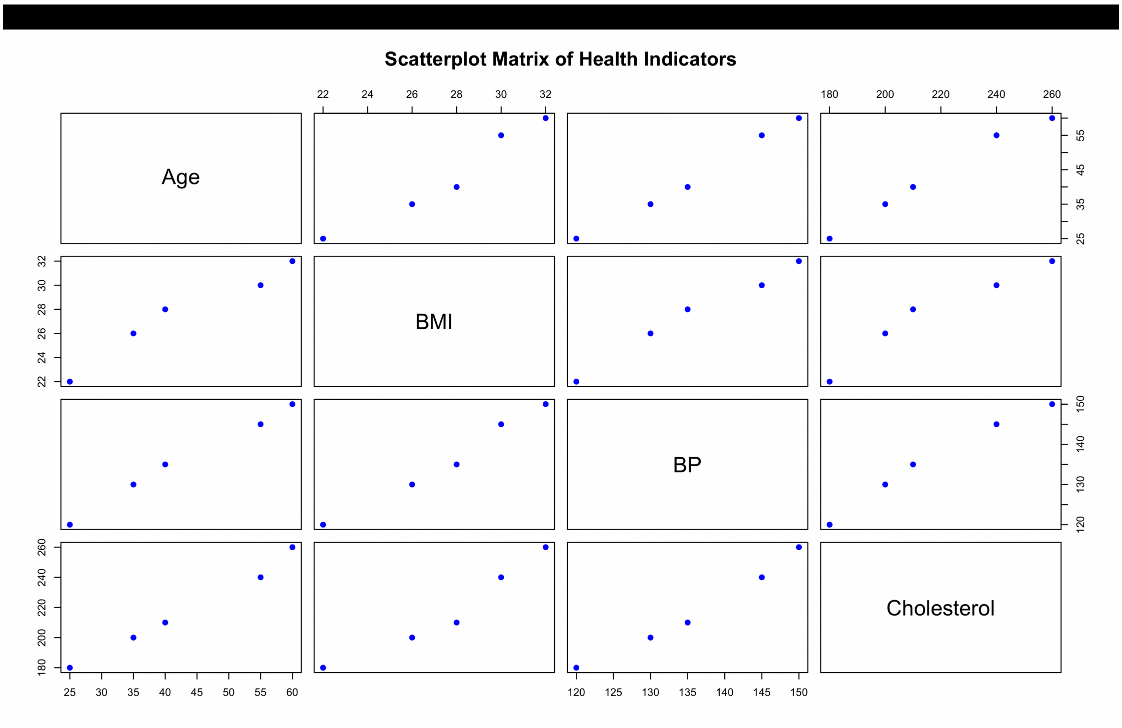
Q1. Scatterplot Matrix for Health Indicators

Program:

```
patient <- data.frame(
  Patient_ID = c("P1","P2","P3","P4","P5"),
  Age = c(25,40,55,35,60),
  BMI = c(22,28,30,26,32),
  BP = c(120,135,145,130,150),
  Cholesterol = c(180,210,240,200,260)
```

```
)

pairs(patient[, c("Age","BMI","BP","Cholesterol")],
      main = "Scatterplot Matrix of Health Indicators",
      pch = 19,
      col = “blue")
```

Output:



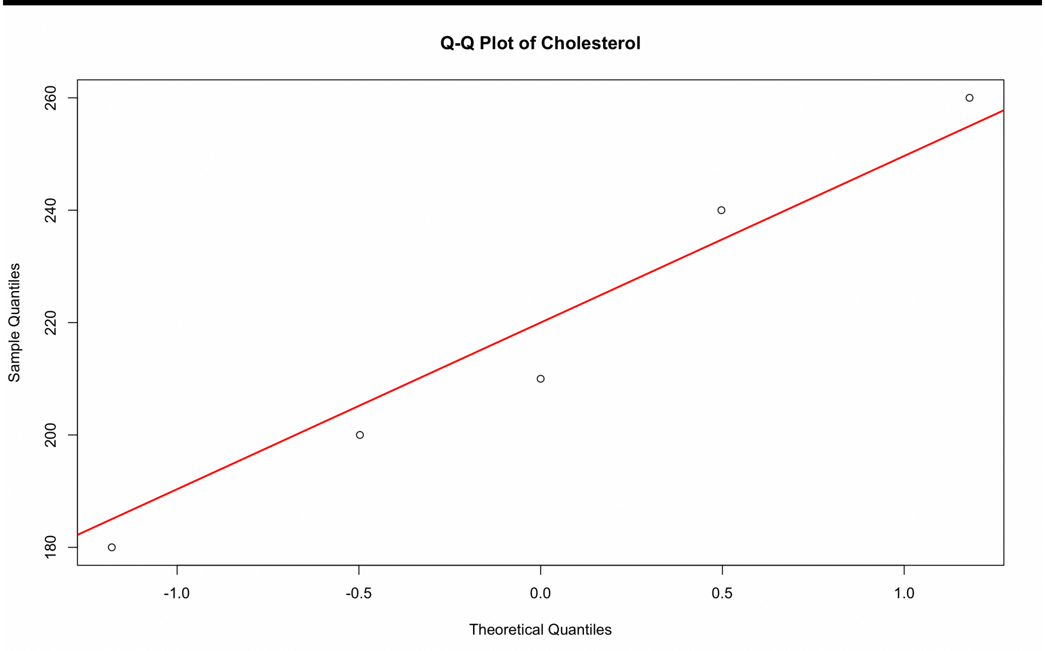
Q2. Q-Q Plot and ECDF for Cholesterol

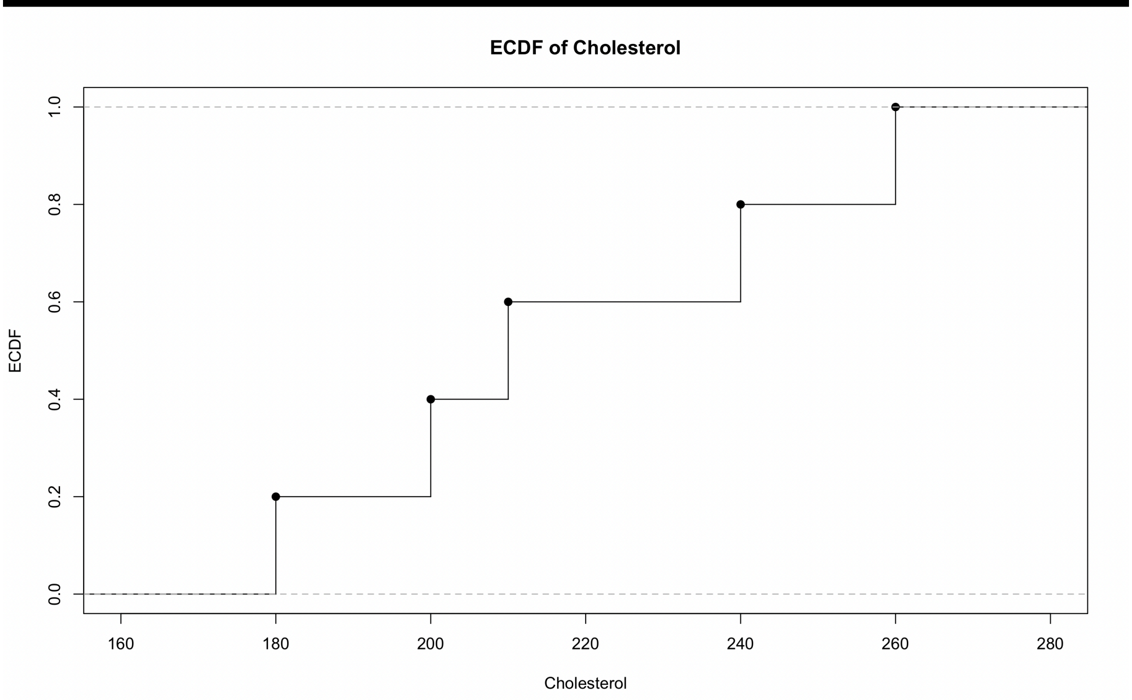
Program:

```
patient <- data.frame(
  Patient_ID = c("P1","P2","P3","P4","P5"),
  Age = c(25,40,55,35,60),
  BMI = c(22,28,30,26,32),
  BP = c(120,135,145,130,150),
  Cholesterol = c(180,210,240,200,260)
)


# Q-Q Plot
qqnorm(patient$Cholesterol,
       main = "Q-Q Plot of Cholesterol")
qqline(patient$Cholesterol,
```

```
      col = "red",
      lwd = 2)

# ECDF
plot(ecdf(patient$Cholesterol),
     main = "ECDF of Cholesterol",
     xlab = "Cholesterol",
     ylab = "ECDF",
     verticals = TRUE,
     do.points = TRUE)
```

Output:



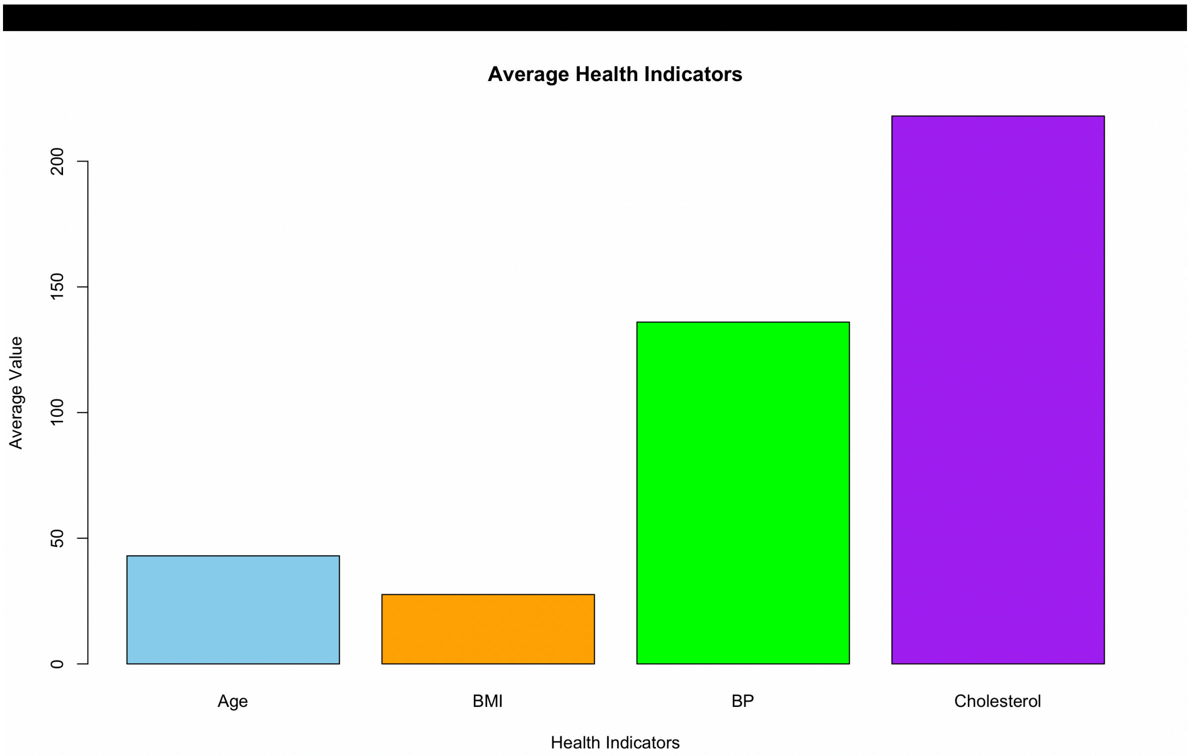


Q3. Bar Chart of Average Health Indicators

Program:

```
patient <- data.frame(
  Patient_ID = c("P1","P2","P3","P4","P5"),
  Age = c(25,40,55,35,60),
  BMI = c(22,28,30,26,32),
  BP = c(120,135,145,130,150),
  Cholesterol = c(180,210,240,200,260)
)


avg_values <- c(
  Age = mean(patient$Age),
  BMI = mean(patient$BMI),
  BP = mean(patient$BP),
  Cholesterol = mean(patient$Cholesterol)
)

barplot(avg_values,
        main = "Average Health Indicators",
        xlab = "Health Indicators",
        ylab = "Average Value",
        col = c("skyblue","orange","green","purple"),
        names.arg = names(avg_values))
```

Output:



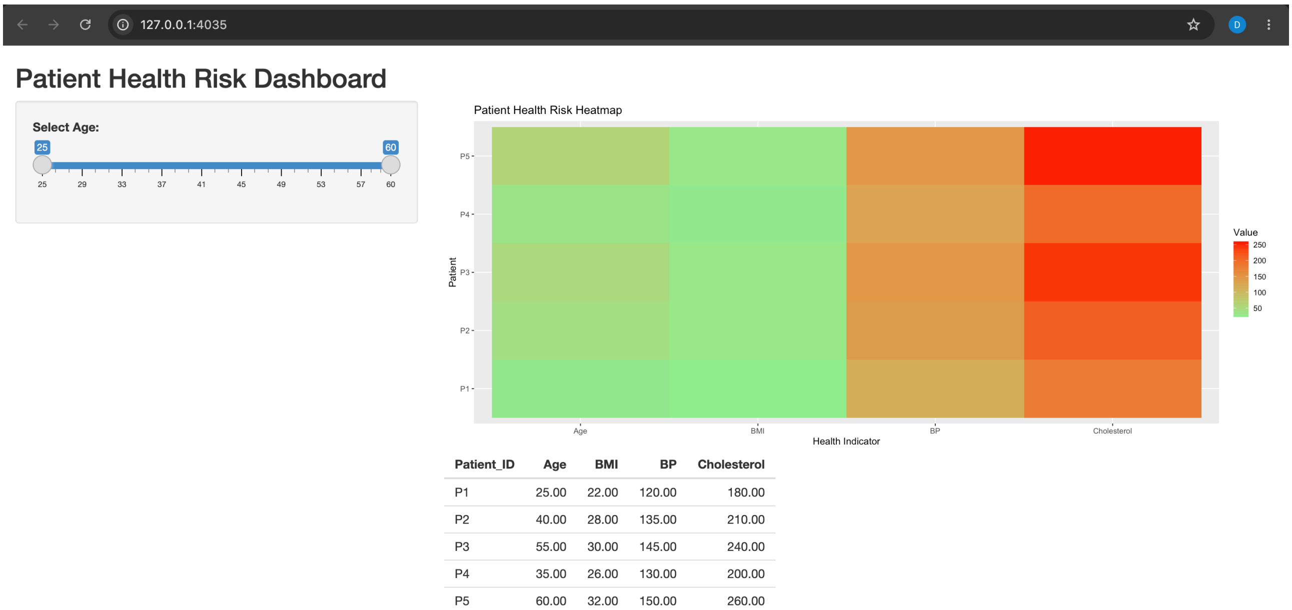
Q4. Interactive Patient Health Risk Dashboard

Program:

```
install.packages("shiny")
install.packages("ggplot2")

library(shiny)
library(ggplot2)

patient <- data.frame(
  Patient_ID = c("P1","P2","P3","P4","P5"),
  Age = c(25,40,55,35,60),
  BMI = c(22,28,30,26,32),
  BP = c(120,135,145,130,150),
  Cholesterol = c(180,210,240,200,260)
)

ui <- fluidPage(

  titlePanel("Patient Health Risk Dashboard"),

  sidebarLayout(
    sidebarPanel(
      sliderInput("age",
                  "Select Age:",
                  min = min(patient$Age),
                  max = max(patient$Age),
                  value = c(min(patient$Age), max(patient$Age)))
    ),

    mainPanel(
      plotOutput("heatmap"),
      tableOutput("data")
    )
  )
)

server <- function(input, output) {

  filtered_data <- reactive({
    patient[patient$Age >= input$age[1] &
              patient$Age <= input$age[2], ]
  })

  output$heatmap <- renderPlot({
```

```
    data <- filtered_data()

    heat_data <- data.frame(
      Patient = rep(data$Patient_ID, 4),
      Indicator = rep(c("Age","BMI","BP","Cholesterol"),
                      each = nrow(data)),
      Value = c(data$Age,
                data$BMI,
                data$BP,
                data$Cholesterol)
    )

    ggplot(heat_data,
           aes(x = Indicator,
               y = Patient,
               fill = Value)) +
      geom_tile() +
      labs(title = "Patient Health Risk Heatmap",
           x = "Health Indicator",
           y = "Patient") +
      scale_fill_gradient(low = "lightgreen",
                          high = "red")
  })

  output$data <- renderTable({
    filtered_data()
  })
}

shinyApp(ui = ui, server = server)
```

Output:



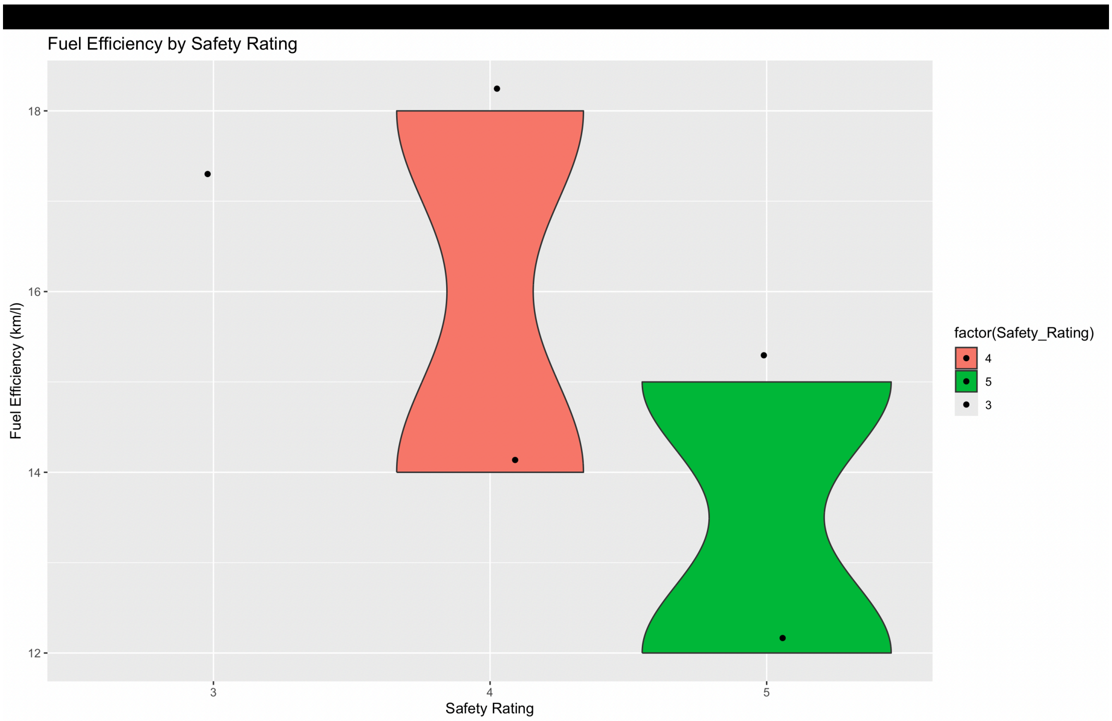
SET 28: Vehicle Performance Analysis

Q1. Violin Plot of Fuel Efficiency by Safety Rating

Program:

```
vehicle <- data.frame(
  Vehicle_ID = c("V1","V2","V3","V4","V5"),
  Engine_Size = c(1.5,2.0,3.0,2.5,1.8),
  Horsepower = c(110,150,250,200,130),
  Fuel_Efficiency = c(18,15,12,14,17),
  Top_Speed = c(180,200,250,220,190),
  Safety_Rating = c(4,5,5,4,3)
)

library(ggplot2)

ggplot(vehicle, aes(x=factor(Safety_Rating),
                    y=Fuel_Efficiency,
                    fill=factor(Safety_Rating))) +
  geom_violin() +
  geom_jitter(width=0.1) +
  labs(title="Fuel Efficiency by Safety Rating",
       x="Safety Rating",
       y="Fuel Efficiency (km/l)”)
```

Output:

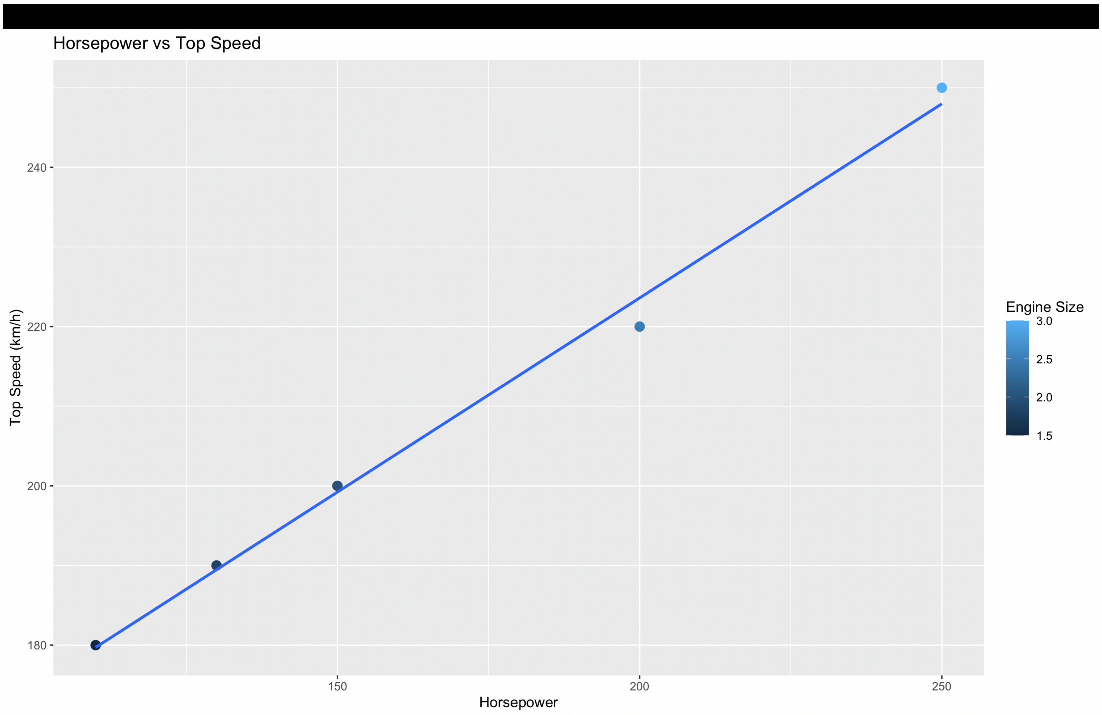


Q2. Horsepower vs Top Speed by Engine Size

Program:

```
vehicle <- data.frame(
  Vehicle_ID = c("V1","V2","V3","V4","V5"),
  Engine_Size = c(1.5,2.0,3.0,2.5,1.8),
  Horsepower = c(110,150,250,200,130),
  Fuel_Efficiency = c(18,15,12,14,17),
  Top_Speed = c(180,200,250,220,190),
  Safety_Rating = c(4,5,5,4,3)
)

library(ggplot2)
ggplot(vehicle, aes(x=Horsepower,
                    y=Top_Speed,
                    color=Engine_Size)) +
  geom_point(size=3) +
  geom_smooth(method="lm", se=FALSE) +
  labs(title="Horsepower vs Top Speed",
       x="Horsepower",
       y="Top Speed (km/h)",
       color="Engine Size”)
```

Output:

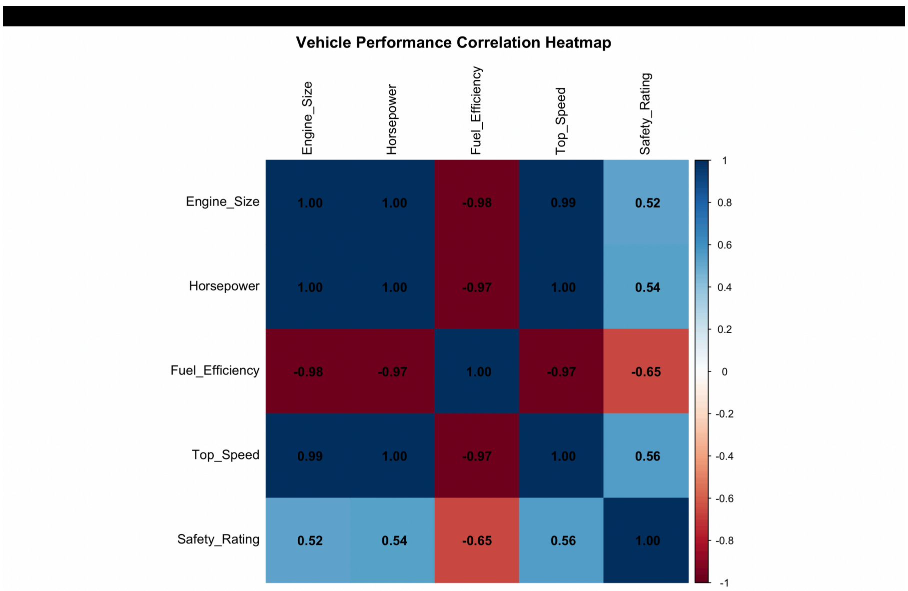


Q3. Correlation Heatmap of Vehicle Performance

Program:

```r
vehicle <- data.frame(
  Vehicle_ID = c("V1","V2","V3","V4","V5"),
  Engine_Size = c(1.5,2.0,3.0,2.5,1.8),
  Horsepower = c(110,150,250,200,130),
  Fuel_Efficiency = c(18,15,12,14,17),
  Top_Speed = c(180,200,250,220,190),
  Safety_Rating = c(4,5,5,4,3)
)

install.packages("corrplot")
library(corrplot)

num_data <- vehicle[, c("Engine_Size",
                        "Horsepower",
                        "Fuel_Efficiency",
                        "Top_Speed",
                        "Safety_Rating")]

cor_matrix <- cor(num_data)

corrplot(cor_matrix,
         method="color",
         addCoef.col="black",
         tl.col="black",
         title="Vehicle Performance Correlation Heatmap",
         mar=c(0,0,2,0))
```

Output:



Q4. Interactive Vehicle Performance Dashboard

Program:

```
library(shiny)
library(ggplot2)

vehicle <- data.frame(
  Vehicle_ID = c("V1","V2","V3","V4","V5"),
  Engine_Size = c(1.5,2.0,3.0,2.5,1.8),
  Horsepower = c(110,150,250,200,130),
  Fuel_Efficiency = c(18,15,12,14,17),
  Top_Speed = c(180,200,250,220,190),
  Safety_Rating = c(4,5,5,4,3)
)

ui <- fluidPage(
  titlePanel("Vehicle Performance Dashboard"),

  sidebarLayout(
    sidebarPanel(
      sliderInput("engine",
                  "Engine Size:",
                  min = min(vehicle$Engine_Size),
                  max = max(vehicle$Engine_Size),
                  value = c(min(vehicle$Engine_Size),
                            max(vehicle$Engine_Size))),

      checkboxGroupInput("safety",
                         "Safety Rating:",
                         choices = sort(unique(vehicle$Safety_Rating)),
                         selected = sort(unique(vehicle$Safety_Rating)))
    ),

    mainPanel(
      plotOutput("bar"),
      plotOutput("scatter")
    )
  )
)

server <- function(input, output) {

  filtered <- reactive({
    vehicle[
      vehicle$Engine_Size >= input$engine[1] &
      vehicle$Engine_Size <= input$engine[2] &
      vehicle$Safety_Rating %in% input$safety, ]
```

```
 })

 output$bar <- renderPlot({
  ggplot(filtered(),
         aes(x = Vehicle_ID,
             y = Fuel_Efficiency,
             fill = factor(Safety_Rating))) +
    geom_bar(stat = "identity") +
    labs(title = "Fuel Efficiency by Vehicle",
         x = "Vehicle",
         y = "Fuel Efficiency (km/l)",
         fill = "Safety Rating")
 })

 output$scatter <- renderPlot({
  ggplot(filtered(),
         aes(x = Horsepower,
             y = Top_Speed,
             color = Engine_Size)) +
    geom_point(size = 4) +
    labs(title = "Horsepower vs Top Speed",
         x = "Horsepower",
         y = "Top Speed (km/h)",
         color = "Engine Size")
 })
}

shinyApp(ui = ui, server = server)
```

Output:

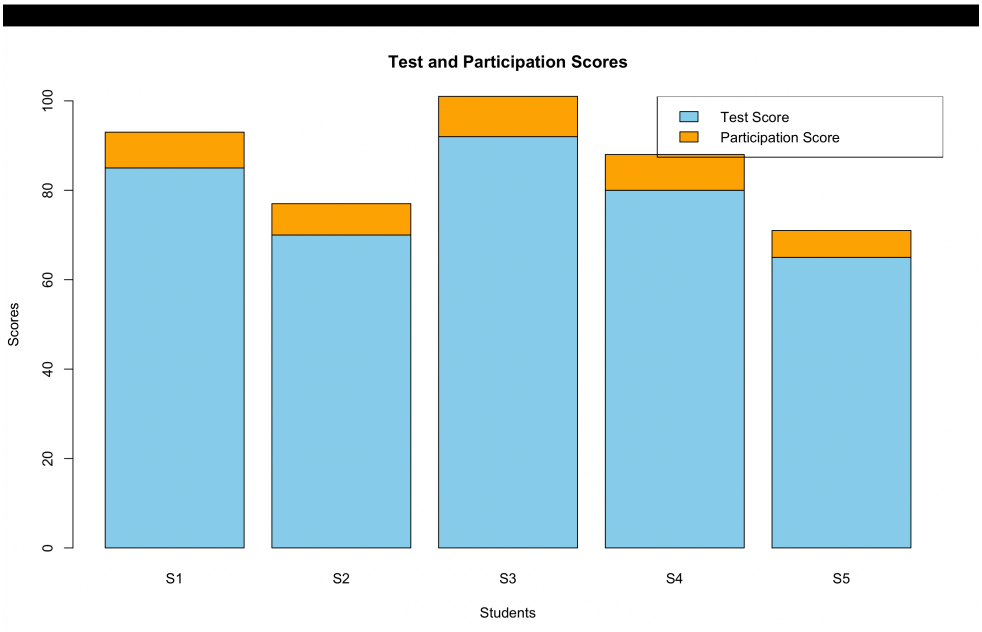
SET 29: Student Academic Performance

Q1. Stacked Area Chart of Test and Participation Scores

Program:

```
student <- data.frame(
  Student_ID = c("S1","S2","S3","S4","S5"),
  Age = c(19,21,20,22,23),
  Study_Hours = c(12,8,15,10,7),
  Attendance = c(90,70,95,85,60),
  Test_Score = c(85,70,92,80,65),
  Participation_Score = c(8,7,9,8,6)
)

scores <- t(as.matrix(student[, c("Test_Score",
                                   "Participation_Score")]))

barplot(scores,
        beside = FALSE,
        col = c("skyblue","orange"),
        names.arg = student$Student_ID,
        main = "Test and Participation Scores",
        xlab = "Students",
        ylab = "Scores")

legend("topleft",
       legend = c("Test Score","Participation Score"),
       fill = c("skyblue","orange"))
```

Output:

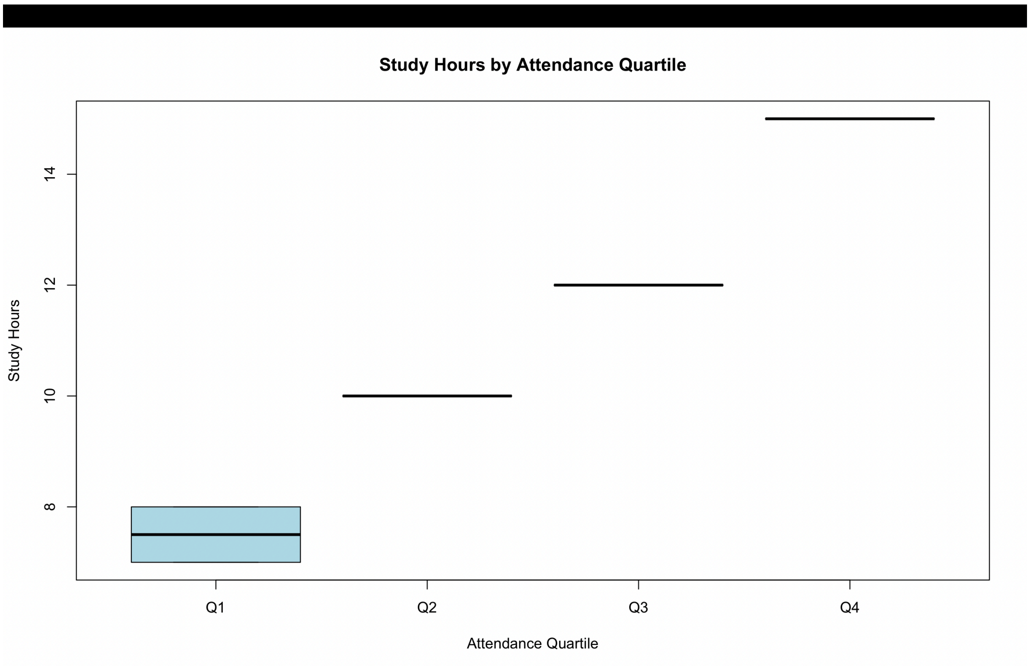


Q2. Boxplot of Study Hours by Attendance Quartiles

Program:

```
student <- data.frame(
  Student_ID = c("S1","S2","S3","S4","S5"),
  Age = c(19,21,20,22,23),
  Study_Hours = c(12,8,15,10,7),
  Attendance = c(90,70,95,85,60),
  Test_Score = c(85,70,92,80,65),
  Participation_Score = c(8,7,9,8,6)
)


student$Attendance_Quartile <- cut(
  student$Attendance,
  breaks = quantile(student$Attendance,
                    probs = c(0,0.25,0.5,0.75,1)),
  include.lowest = TRUE,
  labels = c("Q1","Q2","Q3","Q4")
)

boxplot(Study_Hours ~ Attendance_Quartile,
        data = student,
        col = c("lightblue","lightgreen","orange","pink"),
        main = "Study Hours by Attendance Quartile",
        xlab = "Attendance Quartile",
        ylab = "Study Hours")
```

Output:

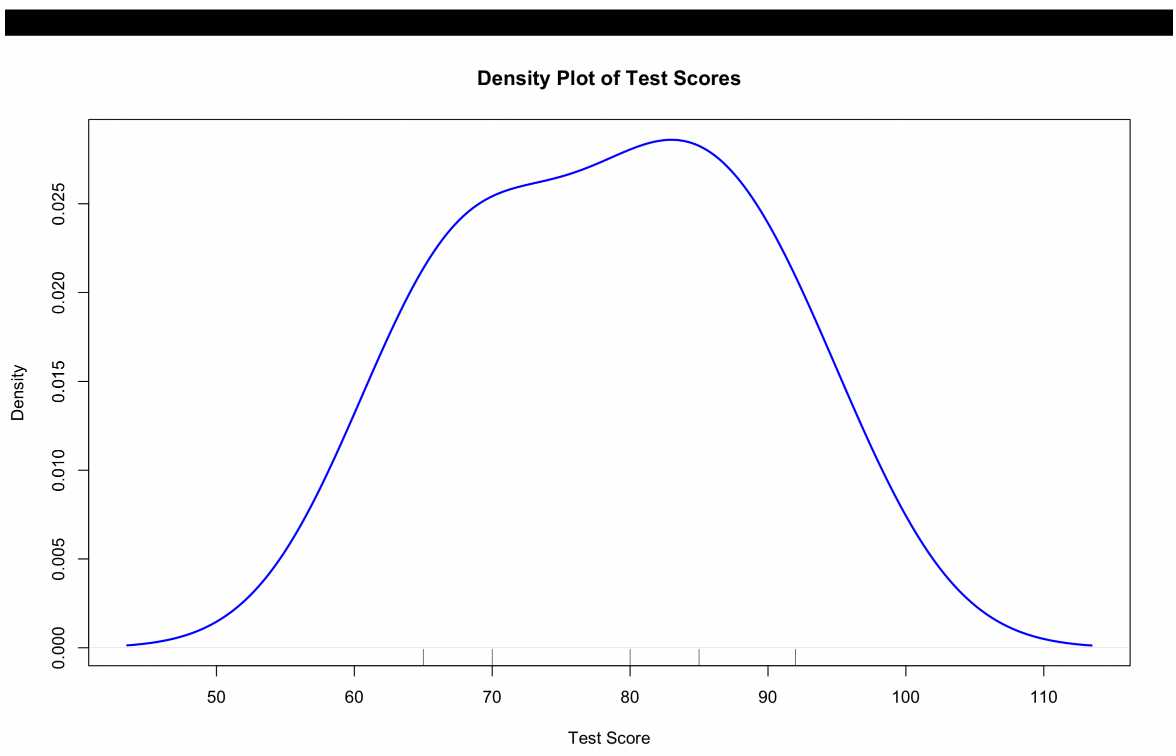


Q3. Density Plot of Test Scores

Program:

```
student <- data.frame(
  Student_ID = c("S1","S2","S3","S4","S5"),
  Age = c(19,21,20,22,23),
  Study_Hours = c(12,8,15,10,7),
  Attendance = c(90,70,95,85,60),
  Test_Score = c(85,70,92,80,65),
  Participation_Score = c(8,7,9,8,6)
)

plot(density(student$Test_Score),
     main = "Density Plot of Test Scores",
     xlab = "Test Score",
     ylab = "Density",
     col = "blue",
     lwd = 2)

rug(student$Test_Score)
```

Output:



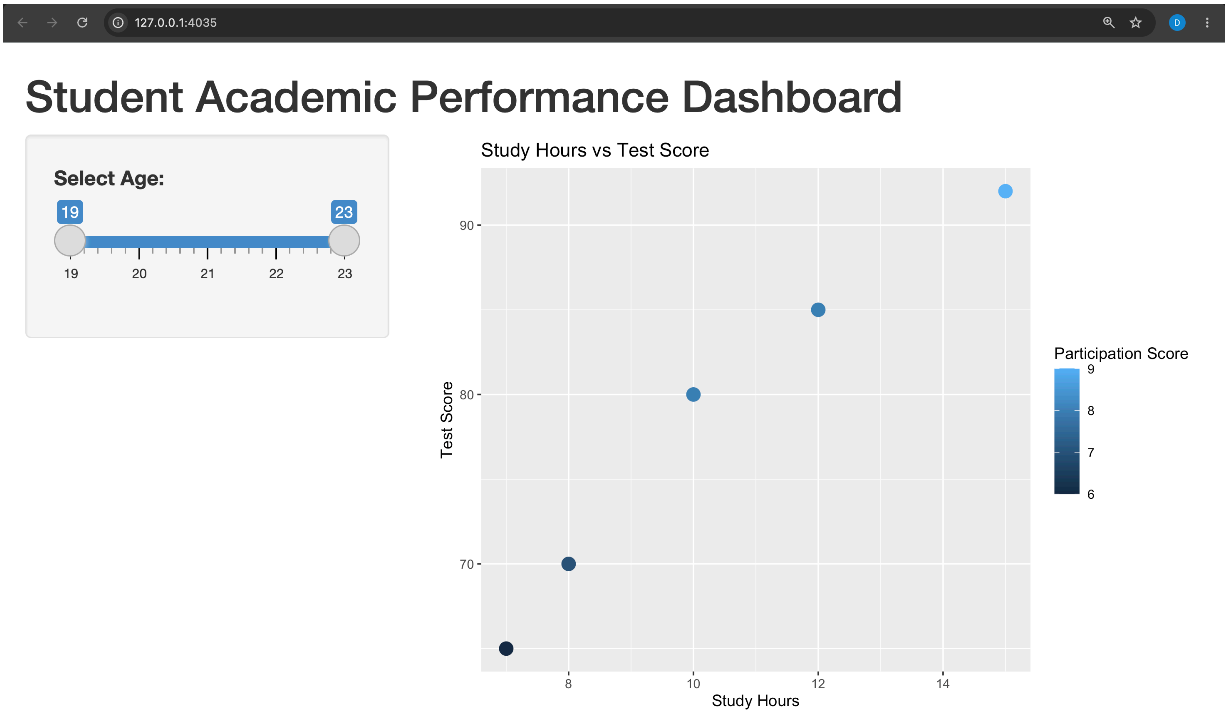
Q4. Interactive Scatterplot of Study Hours and Test Scores

Program:

```
install.packages("shiny")
install.packages("ggplot2")

library(shiny)
library(ggplot2)

ui <- fluidPage(
  titlePanel("Student Academic Performance Dashboard"),

  sidebarLayout(
    sidebarPanel(
      sliderInput("age",
                  "Select Age:",
                  min = min(student$Age),
                  max = max(student$Age),
                  value = c(min(student$Age),
                            max(student$Age)))
    ),

    mainPanel(
      plotOutput("scatter")
    )
  )
)

server <- function(input, output) {

  filtered <- reactive({
    student[student$Age >= input$age[1] &
              student$Age <= input$age[2], ]
  })

  output$scatter <- renderPlot({
    ggplot(filtered(),
           aes(x = Study_Hours,
               y = Test_Score,
               color = Participation_Score)) +
      geom_point(size = 4) +
      labs(title = "Study Hours vs Test Score",
           x = "Study Hours",
           y = "Test Score",
           color = "Participation Score")
  })
}
```

shinyApp(ui = ui, server = server)

Output:



SET 30

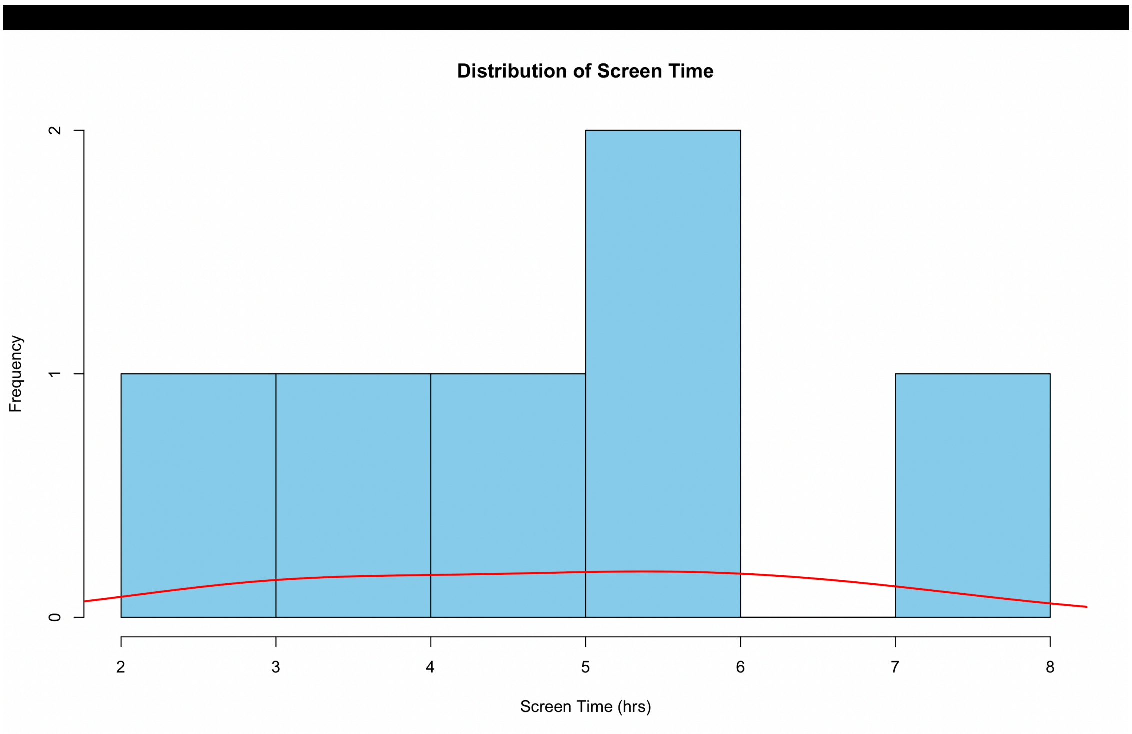
Q1. Histogram and Density Plot of Screen Time

Program:

```
mobile <- data.frame(
  User_ID = c("U01","U02","U03","U04","U05","U06"),
  Gender = c("Male","Female","Male","Female","Male","Female"),
  Age = c(20,22,19,21,23,20),
  Screen_Time = c(4.5,6.0,3.2,7.1,2.8,5.4),
  App_Usage_Count = c(18,25,12,30,10,22),
  Data_Used = c(2.4,3.8,1.6,4.5,1.2,3.1),
  Satisfaction = c(3,5,3,5,2,4),
  Usage_Date = as.Date(c("2025-01-08","2025-01-08",
                         "2025-02-11","2025-02-11",
                         "2025-03-14","2025-03-14"))
)

hist(mobile$Screen_Time,
     main = "Distribution of Screen Time",
```

```
     xlab = "Screen Time (hrs)",
     ylab = "Frequency",
     col = "skyblue",
     border = "black")

lines(density(mobile$Screen_Time),
      col = "red",
      lwd = 2)
```

Output:



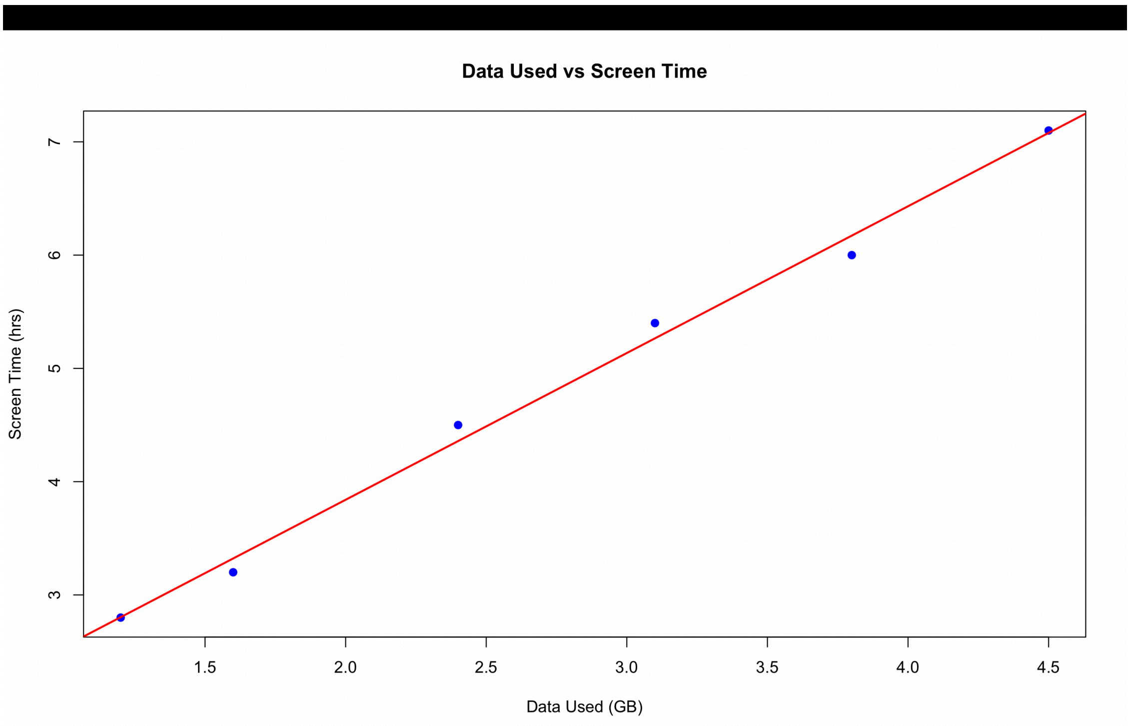
Q2. Data Used vs Screen Time with Correlation

Program:

```
mobile <- data.frame(
  User_ID = c("U01","U02","U03","U04","U05","U06"),
  Gender = c("Male","Female","Male","Female","Male","Female"),
  Age = c(20,22,19,21,23,20),
  Screen_Time = c(4.5,6.0,3.2,7.1,2.8,5.4),
  App_Usage_Count = c(18,25,12,30,10,22),
  Data_Used = c(2.4,3.8,1.6,4.5,1.2,3.1),
  Satisfaction = c(3,5,3,5,2,4),
  Usage_Date = as.Date(c("2025-01-08","2025-01-08",
                        "2025-02-11","2025-02-11",
```

```
                        "2025-03-14","2025-03-14"))
)

plot(mobile$Data_Used,
     mobile$Screen_Time,
     main = "Data Used vs Screen Time",
     xlab = "Data Used (GB)",
     ylab = "Screen Time (hrs)",
     col = "blue",
     pch = 19)

model <- lm(Screen_Time ~ Data_Used, data = mobile)

abline(model,
       col = "red",
       lwd = 2)

cor(mobile$Data_Used,
    mobile$Screen_Time)
```

Output:



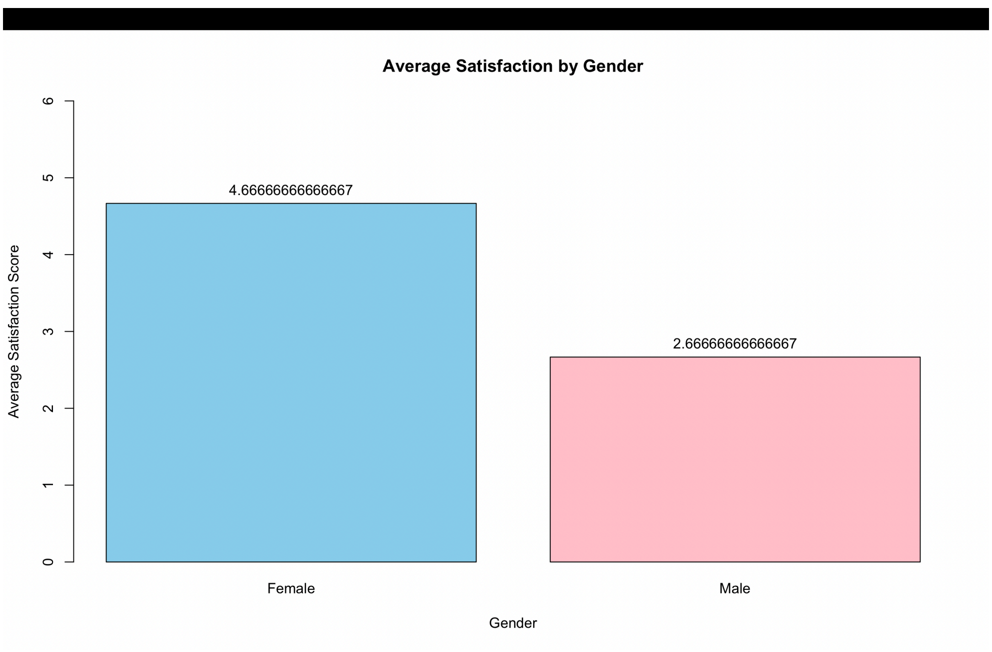
Q3. Average Satisfaction Score by Gender

Program:

```
mobile <- data.frame(
  User_ID = c("U01","U02","U03","U04","U05","U06"),
```

```
  Gender = c("Male","Female","Male","Female","Male","Female"),
  Age = c(20,22,19,21,23,20),
  Screen_Time = c(4.5,6.0,3.2,7.1,2.8,5.4),
  App_Usage_Count = c(18,25,12,30,10,22),
  Data_Used = c(2.4,3.8,1.6,4.5,1.2,3.1),
  Satisfaction = c(3,5,3,5,2,4),
  Usage_Date = as.Date(c("2025-01-08","2025-01-08",
                         "2025-02-11","2025-02-11",
                         "2025-03-14","2025-03-14"))
)
avg_satisfaction <- aggregate(
  Satisfaction ~ Gender,
  data = mobile,
  FUN = mean
)

print(avg_satisfaction)

barplot(avg_satisfaction$Satisfaction,
        names.arg = avg_satisfaction$Gender,
        main = "Average Satisfaction by Gender",
        xlab = "Gender",
        ylab = "Average Satisfaction Score",
        col = c("skyblue", "pink"),
        ylim = c(0, 6))

text(x = c(0.7, 1.9),
     y = avg_satisfaction$Satisfaction,
     labels = avg_satisfaction$Satisfaction,
     pos = 3)
```

Output:



Q4. Mobile App Usage Dashboard

Program:

```
install.packages("shiny")
install.packages("ggplot2")

library(shiny)
library(ggplot2)

ui <- fluidPage(

  titlePanel("Mobile App Usage Dashboard"),

  sidebarLayout(
    sidebarPanel(
      selectInput("gender",
                  "Gender:",
                  choices = c("All",
                              unique(mobile$Gender))),

      dateRangeInput("date",
                     "Usage Date:",
                     start = min(mobile$Usage_Date),
                     end = max(mobile$Usage_Date))
    ),

    mainPanel(
      plotOutput("bubble"),
      plotOutput("trend")
    )
  )
)

server <- function(input, output) {

  filtered <- reactive({
    data <- mobile[
      mobile$Usage_Date >= input$date[1] &
      mobile$Usage_Date <= input$date[2], ]

    if (input$gender != "All") {
      data <- data[data$Gender == input$gender, ]
    }

    data
  })
```

```
  output$bubble <- renderPlot({
    ggplot(filtered(),
           aes(x = Screen_Time,
               y = Data_Used,
               color = Gender,
               size = App_Usage_Count)) +
      geom_point(alpha = 0.8) +
      labs(title = "Screen Time vs Data Used",
           x = "Screen Time (hrs)",
           y = "Data Used (GB)",
           color = "Gender",
           size = "App Usage Count")
  })

  output$trend <- renderPlot({
    ggplot(filtered(),
           aes(x = Usage_Date,
               y = Screen_Time,
               color = Gender)) +
      geom_line() +
      geom_point(size = 3) +
      labs(title = "Usage Trend Over Time",
           x = "Usage Date",
           y = "Screen Time (hrs)",
           color = "Gender")
  })
}

shinyApp(ui = ui, server = server)
```

Output: